

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Natural Sciences (NSc)

SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2024-2025

DURATION: 2 HOURS

FULL MARKS: 160

MATH 4141: Geometry and Differential Calculus

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 4 (four) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. (a) Check whether the equation $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$ represents a pair of straight line or not. If so, find the equation of the lines, their point of intersection, angles between the lines, the equation of the bisectors of the angles between them, and finally sketch the lines showing the necessary information. [10]
(CO1)
(PO1)

- (b) Show that the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$ and $lx + my = 1$ is right angled if $(a+b)(al^2 + 2hlm + bm^2) = 0$. [10]
(CO2)
(PO2)

- (c) If the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines equidistant from the origin, then prove that [20]
(CO1)
(PO2)

(i). $f^4 - g^4 = c(bf^2 - ag^2)$.

(ii). $h(g^2 - f^2) = gf(a - b)$.

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = a(bc - f^2) - h(bc - gf) + g(hf - bg)$$

2. (a) Define direction cosines and direction ratios of a straight line. Find the projection of the joining of two points $(-1, -1, 3)$ and $(2, 0, 1)$ on the line through the points $(-7, 5, 3)$ and $(2, 6, 8)$. [10]
(CO1)
(PO1)

- (b) Find the perpendicular distance of the point $(-2, 3, 4)$ from the straight line through the point $(-1, 3, 2)$ whose direction ratios are $12, 3, -4$. [10]
(CO2)
(PO2)

- (c) Prove that the two straight lines whose direction cosines are given by the two relations $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are perpendicular or parallel [10]
(CO3)
(PO2)

according as $a^2(v+w) + b^2(w+u) + c^2(u+v) = 0$ or $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$.

3. (a) (i). Find the equation of the plane passing through the line of intersection of the planes $2x - y = 0$ and $y - 3z = 0$, and is perpendicular to the plane $4x + 5y - 3z + 7 = 0$. [20]
(CO1)
(PO2)
- (ii). Find the equations of the two planes passing through the points $(0, 4, -3)$ and $(6, -4, 3)$, if the sum of their intercepts from the coordinate axes is zero.
- (b) Find the magnitude and the equation of the shortest distance between the lines $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$ and $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$. Also, find the coordinates of the points where the shortest distance meets the given lines. [10]
(CO2)
(PO2)
- (c) Find the equation of the line through the point $(1, 2, 3)$ and is parallel to the two planes $x - y + 2z - 5 = 0$ and $3x + y + z - 6 = 0$. [10]
(CO2)
(PO2)
4. (a) (i). State and prove Euler's Theorem for a homogeneous function of two variables. Hence, verify Euler's Theorem for the function $u(x, y) = \frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{5}} + y^{\frac{1}{5}}}$. [20]
(CO3)
(PO1)
- (ii). If $\sin u = \frac{x^3 + y^3}{x + y}$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \tan u$.
- (b) What is meant by envelope of a family of curves? Find the envelope of the family of straight line $y = mx + \sqrt{a^2 m^2 + b^2}$, where m is a parameter. [10]
(CO1)
(PO1)
- (c) (i). Find the radius of curvature at the point $(2a, 4a)$ of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$. [10]
(CO2)
(PO2)
- (ii). Find the n th Maclaurin polynomial for $\sin x$.

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